



Introduction

Automated Climate Control System



Agenda

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Current Problem

- **Temperature Swings:**

Daytime heat and nighttime cold create large temperature changes. This stresses plants, slows their growth, and also disrupts controlled bacterial environments, leading to inconsistent or unwanted bacterial activity.

- **Drying Out of soil:**

Watering system failures cause under-watering or missed watering cycles, leading to dry soil, unhealthy plants, and unfavorable moisture conditions that affect beneficial bacterial survival in the growing medium.

- **Mold and Disease Growth:**

High humidity and stagnant moisture allow mold, plant diseases, and harmful to spread inside the greenhouse, increasing crop damage and contamination risks.



Our Idea

We are building a Automated Climate Control System that automatically controls and maintains the environmental conditions using sensors and a microcontroller.

How it works:

- Sensors measure temperature, humidity, soil moisture, and light
- **A controller analyzes the data**
- **Devices respond automatically:**
 - Fan for temperature & humidity
 - Pump for watering
 - Lights for plant and bacterial growth

To create a greenhouse that:

Thinks on its own
Protects crops
Saves water and energy

Our Solution

- **Automatic Temperature Control**

The system maintains ideal temperature by removing hot air when it's too warm and activating heating or reducing airflow when it's cold.

- **Real-Time Soil Moisture Control**

Soil moisture sensors ensure presence of right amount of water in soil. The pump runs only when the soil is dry.

- **Mold Prevention System**

Humidity, airflow, and watering are continuously adjusted to keep the area disease free.

- **Light Stress**

Unstable sunlight triggers premature "bolting," where plants seed instead of growing and in case of bacteria they can suffer DNA damage or metabolic disruption.

- **Data Blindness**

When problems cannot be immediately identified, the system provides historical data analysis to identify root causes and support better decision making

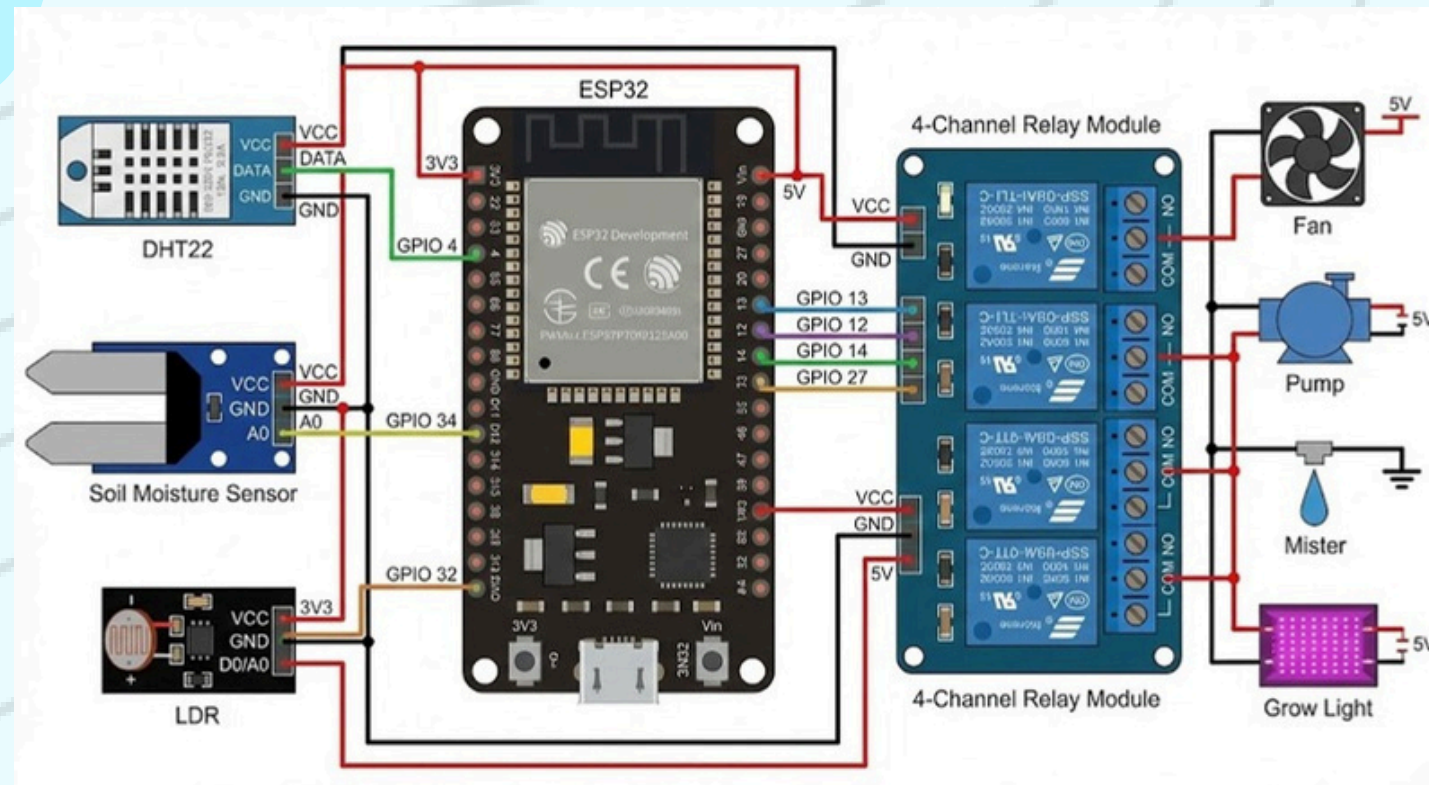
- **Light Stress**

Running fans and lights 24/7 without sensors inflates bills and wears out hardware. Automation slashes costs by using equipment when only required.

Features

- **Automated Climate Control:**
 - Cooling: Fan activates automatically if the temperature exceeds 31°C.
 - Humidification: Mister activates if ambient humidity drops below 45%.
 - Smart Irrigation: Pump triggers only when soil moisture falls below the given parameter, ensuring water conservation.
 - Intelligent Lighting: LED grow lights activate when sunlight level drops below set parameter.
- **Real-Time Remote Monitoring:** Users can view live data and system status from any device via a secure web dashboard.
- **Secure Access:** Multi-user authentication (Login system) protects the dashboard from unauthorized access.
- **Anti-Disease Presets:** Maintain certain climatic condition presets to prevent certain diseases in plants caused due to sustained temperature or humidity conditions.
- **Manual Control:** Modify all the parameters directly from the web app. And have manual control over all the actuators.
- **Growth Optimization:** Use the historical data in MongoDB to determine the "Optimal Growth Profile" for specific plants and variants of bacteria, automatically adjusting light and water according to it.
- **Exportable Data Logs:** Add a feature to download sensor history as a CSV or Excel file for offline research and documentation.

Hardware Components

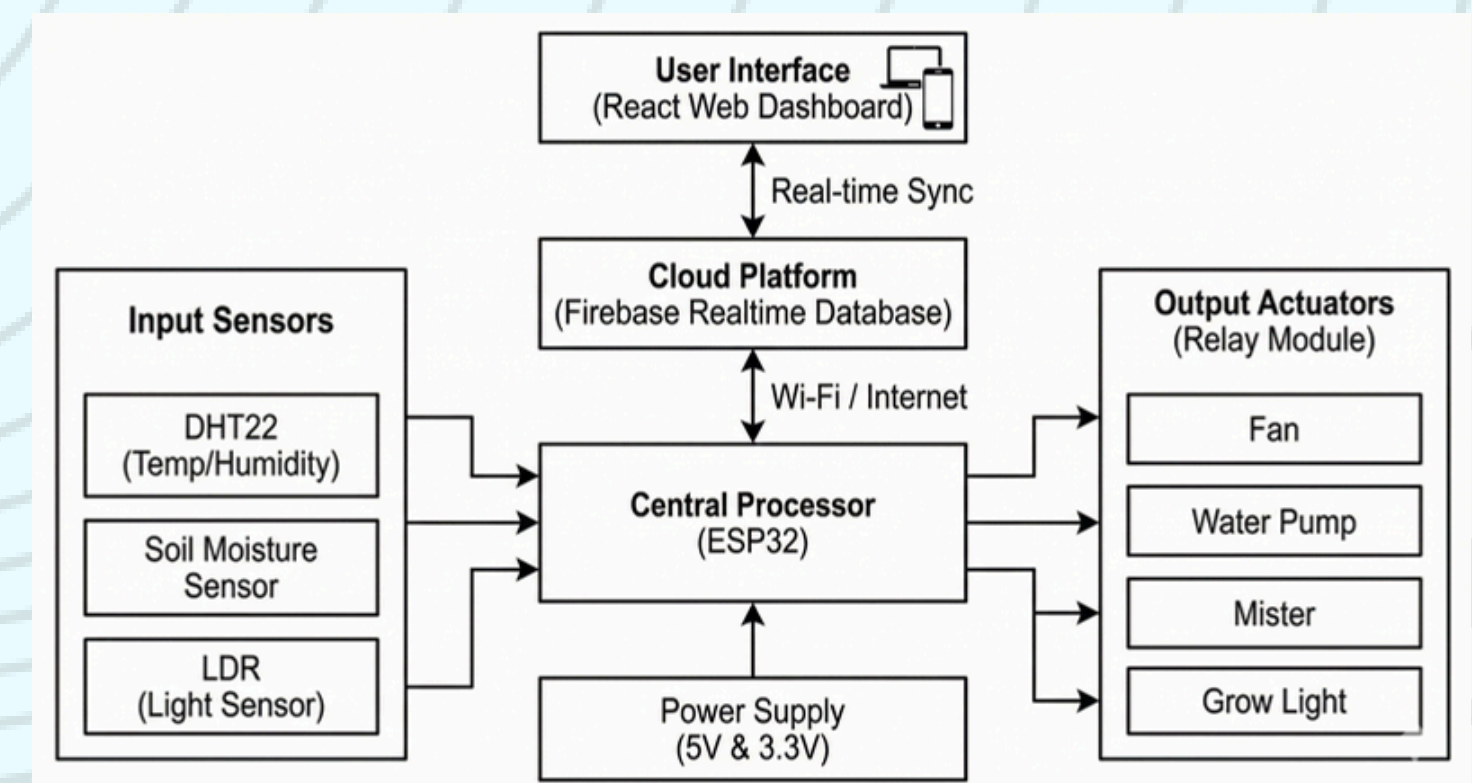


Circuit Diagram

- **Controller:** ESP32 Microcontroller – Acts as the brain of the system, handling sensor data processing and WiFi connectivity.
- **Input Sensors:**
 - DHT22 Sensor: Monitors ambient air temperature and humidity.
 - Soil Moisture Sensor: Measures the water content in the soil.
 - LDR (Light Dependent Resistor): Detects sunlight levels.

• Output Actuators:

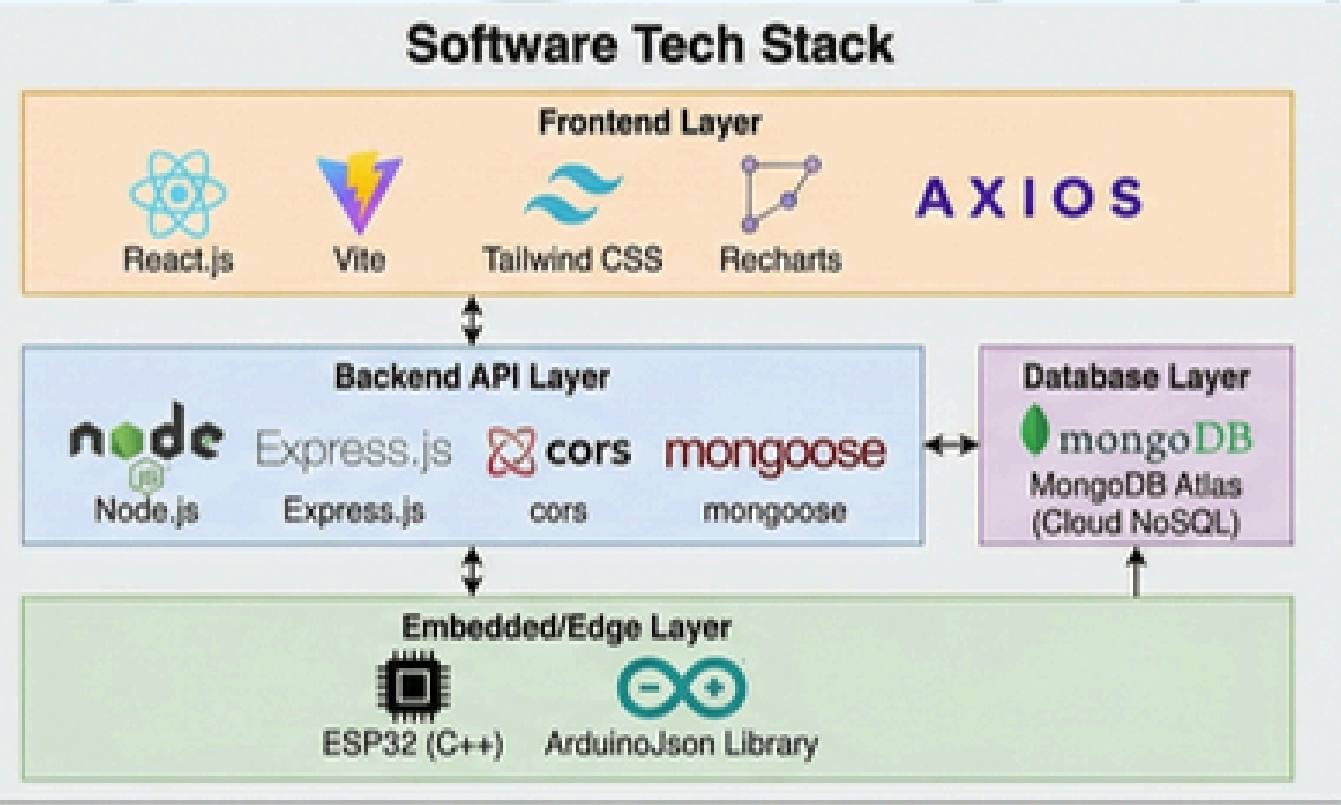
- 4-Channel Relay Module: Switches high-power devices (Fan, Pump, Mister, Grow Light) ON/OFF based on sensor data.
- Submersible Water Pump: For automated drip irrigation.
- DC Cooling Fan: Regulates temperature.
- UV LED Strip: Provides Light during cloud cover.
- Ultrasonic Humidifier: Regulates air humidity.



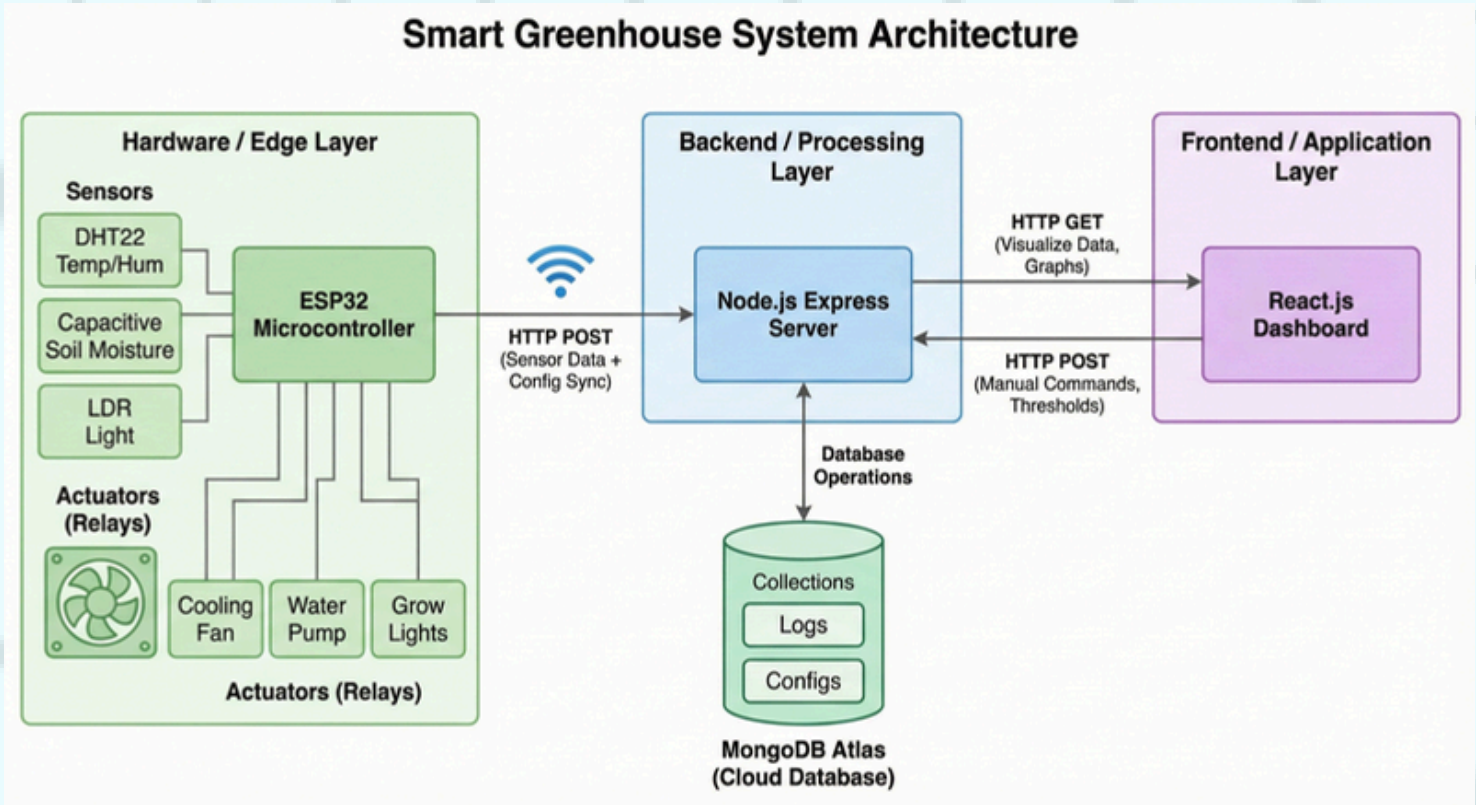
Block Diagram

Software

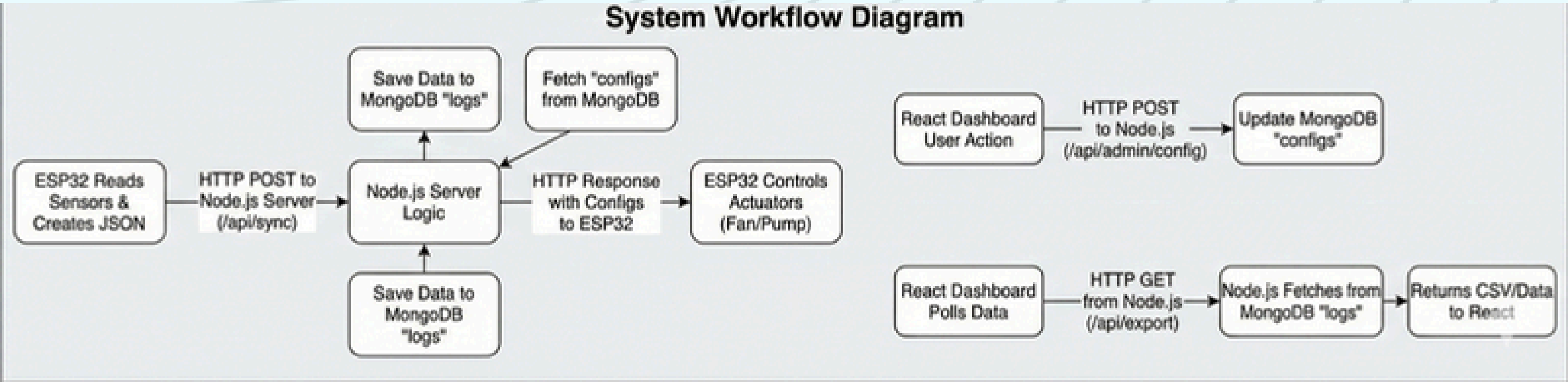
Tech Stack



Architecture Diagram



System Workflow Diagram



Workflow Diagram

WebApp Screenshots



Automated Climate Control System

Sign in to access your dashboard

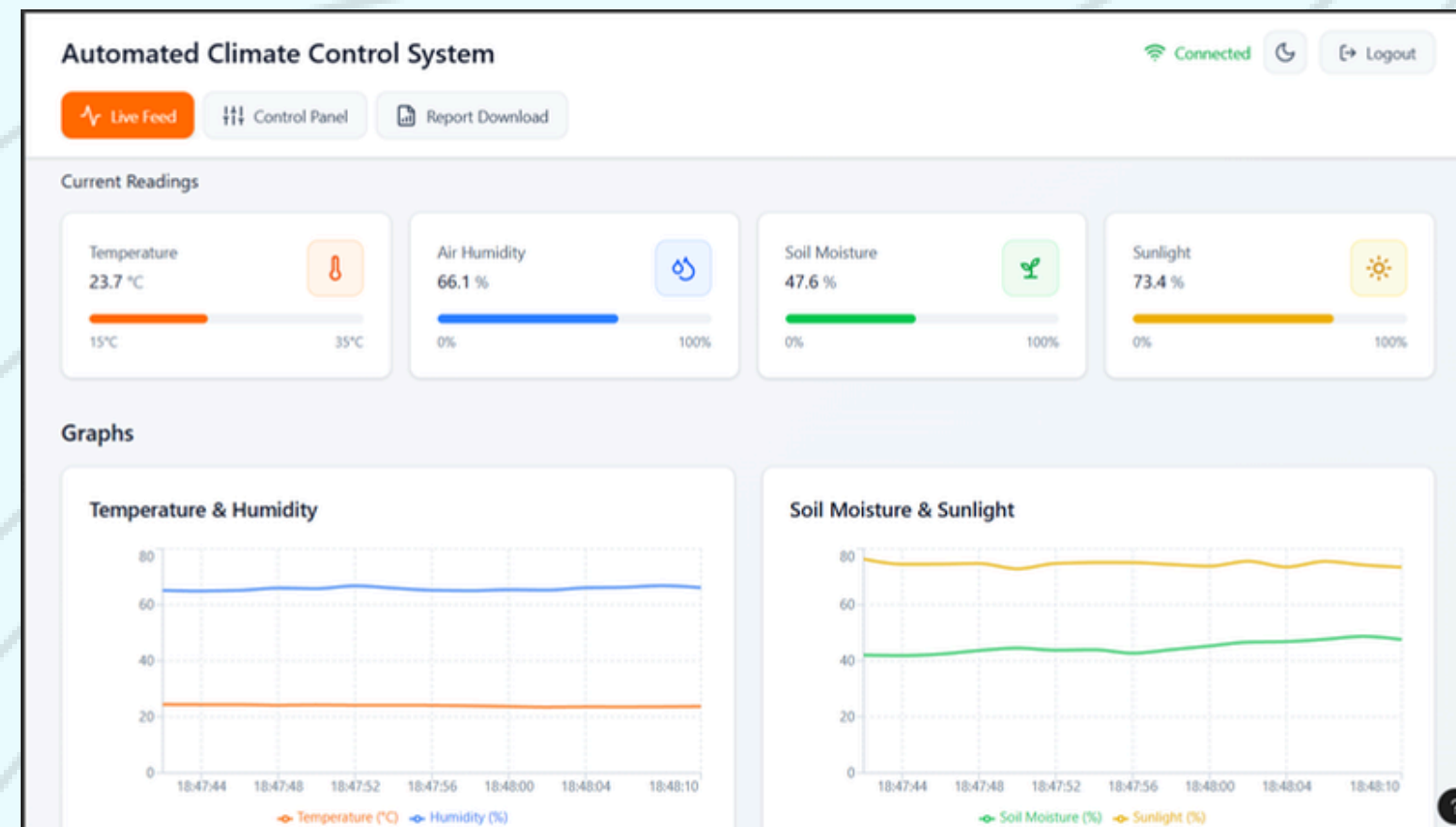
Username

 Enter your username

Password

 Enter your password

Sign In



Automated Climate Control System

Connected Logout

Live Feed Control Panel Report Download

Manual Device Control

Fan
ON
Turn Off

Water Pump
OFF
Turn On

Humidifier
ON
Turn Off

UV Lights
OFF
Turn On

Target Parameter Settings

Parameter Settings

Save Settings

Temperature
25°C
15°C 35°C
25

Air Humidity
60%
30% 90%
60

Soil Moisture
50%
0% 100%
50

Sunlight
70%
0% 100%
70

Note: These parameters will be sent to your ESP32 device when you click "Save Settings". The device will attempt to maintain these target values automatically.

Automated Climate Control System

Connected Logout

Live Feed Control Panel Report Download

Export Sensor Data

Download Statistics

Select Date Range

Today
24 hourly readings

Last 7 Days
~168 hourly readings

Last 30 Days
~720 hourly readings

Export Format

Excel Spreadsheet
.xlsx format

Includes all sensor readings

Summary statistics sheet

Ready for analysis

Download Excel File

Note: The Excel file will contain two sheets: "Sensor Data" with detailed readings and "Summary" with statistical averages. Data is generated from your ESP32 historical records.

Competitors & Their Features

Our competitors in the market are:

- **Netafim IoT Irrigation & Climate Control -**

Real-time soil moisture monitoring, automated water scheduling, cloud analytics, smart irrigation optimization.

- **Monnit Remote Monitoring System -**

Wireless sensors for temperature, humidity, soil moisture with alerts (email/text), cloud dashboard for remote access.

- **IoX-Connect IoT Greenhouse Solutions -**

Sensors + automated irrigation, lighting, climate controls, scalable platform for small to large greenhouses.

- **Priva / Climate Corporation / Trimble greenhouse systems -**

Full greenhouse climate automation — environmental sensing, irrigation automation, data-driven insights, analytics tools.



Our Edge Over Them

Cost-Effectiveness

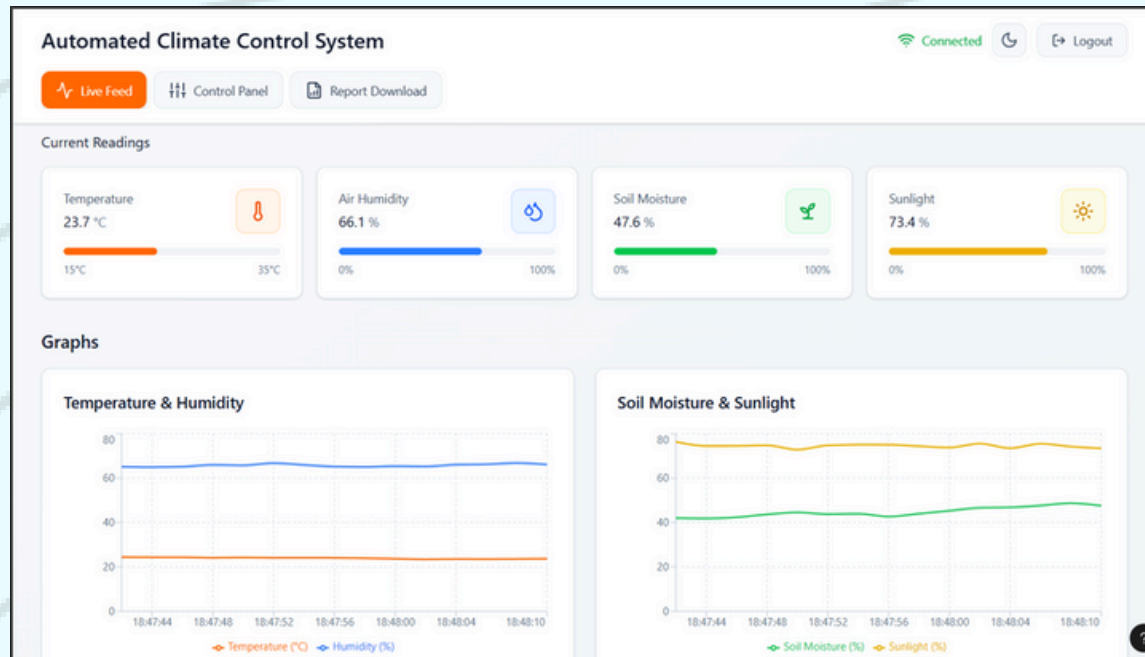
- a. Commercial systems are expensive. our ESP32 + open dashboard can be much cheaper.
- b. DIY products are low cost but often lack automation or scalable web interface.



Custom Web Dashboard

Building a custom website gives:

- a. Full design control that is tailored to greenhouse needs
- b. No dependency on third-party dashboards
- c. Ability to add alerts, history graphs, multi-user access.



Modularity & Expandability

We can add Additional sensors (CO₂, pH, EC, etc.)

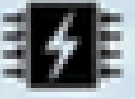
- a. Actuators (fans, heaters, irrigation valves)
- b. Predictive models (alerts based on trends)



Feasibility

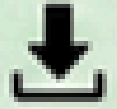


1. TECHNICAL FEASIBILITY

-  **ESP32:** Sufficient processing power, built-in Wi-Fi, multiple GPIO pins.
-  **Sensors:** Temperature, humidity, soil moisture, and light readily available and easy to interface.
-  **Relay modules:** Enable safe control of high-power devices.
-  **Open-source libraries** and documentation support rapid development.
-  **Implementation** using commonly available tools: Arduino IDE and basic web technologies.

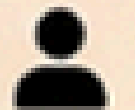
ESP32
SMART
SYSTEM
CORE

2. ECONOMIC FEASIBILITY

-  **Components:** Low-cost and easily affordable.
-  No expensive proprietary hardware or paid software required.
-  **Open-source platforms:** Eliminate licensing costs.



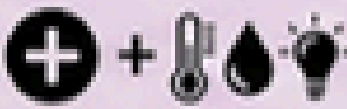
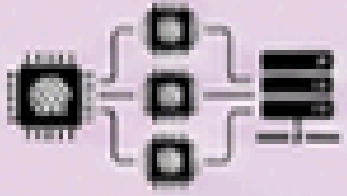
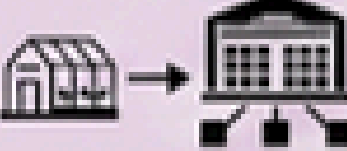
OPERATIONAL FEASIBILITY

-  Minimal user training is required.
-  Automatic monitoring reduces manual intervention.
-  Web-based dashboard: Provides easy access from anywhere.

Conclusion: The system is user-friendly and operationally feasible.



4. SCALABILITY FEASIBILITY

-  Additional sensors and actuators can be added easily.
-  Multiple ESP32 units can be connected to a single server.
-  System scales from small greenhouses to commercial setups.

Impact & Benefits

1. Agricultural Impact:

- Enables real-time monitoring of greenhouse conditions.
- Helps maintain optimal environmental conditions for crops.
- Reduces crop stress and improves yield quality.
- Prevents damage due to extreme temperature or moisture levels.



2. Environmental Benefits:

- Optimized irrigation reduces water wastage.
- Energy-efficient usage of lighting and cooling systems.
- Supports sustainable and smart farming practices.
- Minimizes overuse of resources.



3. Economic Benefits:

- Reduces labor costs through automation.
- Prevents crop losses caused by delayed response.
- Improves productivity with minimal investment.



4. Social Impact:

- Supports small and medium-scale farmers.
- Encourages adoption of smart agriculture technologies.
- Promotes food security and efficient farming methods.

Upscaling

Multiple Product Variants (Tiered Model):

Basic Model- Our complete hardware system + Live Feed View on the Web App.

Pro Model- Basic model + Manual Control + Higher Accuracy.

Ultra Model- Standard model + option download past parameters in an excel sheet, .csv format file + Highest accuracy + future AI features.

Low-Cost Manufacturing Strategy using:

- ESP 32, common sensors and open source software
- PCB instead of breadboard and doing Bulk component purchase.

Target Multiple Market Segments like:

- Small greenhouse owners
- Nursery farms
- Home gardeners
- Research labs
- Educational institutions

• Branding & Trust building by Providing Customers with :

- Documentation
- Warranty
- Demo videos

Finances

We have using a tiered model system with 3 models Basic, Pro and Ultra model.

1. Basic model

- Reliable performance with average accuracy
- Hardware cost: ~₹2,400
- Basic web platform for real-time monitoring
- Final cost: ~₹3,500

2. Pro Model

- Good accuracy and enhanced reliability
- Hardware cost: ~₹2,800
- Real-time monitoring as well as manual control
- Final cost: ~₹4,700

3. Ultra model

- Very high accuracy and superior reliability
- Hardware cost: ~₹3,700
- Advanced web platform access with Manual parameter control
- Downloadable historical data (Excel & CSV)
- Future AI features
- Final cost: ~₹5,500

Average profit margin here is 35%, also the cost of hardware components is expected to reduce when purchased in bulk.

Future Scope

- Camera to monitor and record the plant/bacteria growth.
- ML model to predict the most suitable climate conditions from past records for better results.
- Use AI to analyze which environmental "set-points" led to the fastest growth or the most stable bacterial cultures, allowing the system to suggest optimal parameters automatically.
- Use image processing to identify early signs of leaf yellowing, mold, or contamination in a lab setting, triggering an automatic alert or quarantine protocol.
- Make custom setups according to the customer's special requirements.
- Integrate CO₂ sensors for high-yield urban farming or pH/EC sensors for hydroponics and microbiology labs to provide a more complete environmental control suite.



Thank You

Byte Sized Brains:

Rushil Shitut

Khushboo Choudhary

Shreya Thakur

